Ch-9

When a solid ~~component~~ is mixed with a liquid, it forms a solution. A solution is a a mixture of solvent (liquid) & solute (solid). Eg- Sugar solution.

When a solution is made by mixing 2 liquids, the one, in the larger quantity is considered as solvent & the one in the lesser quantity is considered to be the solute.

Chashni (Sugar syrup) has more sugar than water, but still water is the solvent.

The solution, in which more solute can be added at a given temp., is called an unsaturated solution. But, when the solute stops dissolving at that temp., the solution becomes a saturated solution.

Concentration is the amount of solute in a fixed quantity of solvent.

Depending on concentration, a solution can dilute (less solute) or concentrated (more solute)

Solubility refers to the ~~a~~ amount of solute that can be dissolved in a solvent at a given temp. Solubility increases with temp.

Water at 70°C will have more solubility than water at 50°C. At 50°C, water will have more solubility ~~that~~ than at 20°C. Cold water has less solubility than hot water.

Ancient India medical practices like Ayurveda used drug formulation for treament. Hydro-alcoholic extracts of herb, were used for drug formulation. Water, oil, ghee & milk were used as solvents for drugs.

Gases can also dissolve in water. Oxygen dissolves in water to a small extent only but is actually very important to marine life. Solubility of gases decreases as the temp. increases.

A wooden stick & an iron rod might be of same shape, size & weight (mass) but still iron rod feels heavy. This is because of density. ~~Densit~~

Density (of both liquid & obj), shape, ~~mass~~ etc reason decide whether something floats or not.

Density can be ~~diffe~~ defined as mass ~~pe~~ present in every unit volume. Formula = $\boxed{\dfrac{\text{Mass}}{\text{Volume}}}$
Unit for density is Kg/m^3 or g/m^3

1 ml of water roughly equals to 1 g. ~~a~~
Eg - Block is 27 g ~~cm~~ & $10 cm^3$
So, $D = \frac{27}{10} = 2.7 \, g/cm^3$

So, this block is 2.7 times denser than water.

Volume measurement

- Liquid - Pour the liquid in a measuring cylinder & read the reading from meniscus. For larger quantities, beakers can be used but they might not be that accurate.

- Regular solids - Eg: Cube
  - Just find its length, width, ~~hig~~ height & use this formula
  - $L \times W \times H$

- Irregular solids - ~~Put~~ Fill ~~&~~ a measuring cylinder (or beaker) with water till it is half full. Then, put that object into the water. You will get a new measurement. Now, subtract the new value with old value, & voila! You have the volume of that solid. This process is based on Archimedes Principle.

☆ For irregular solids, the volume first comes in ml which can be written as an equivalent of cm^3.

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Earth is made up of many layers & as you go deeper, density, pressure & heat increases.

In ancient times, bamboo was used to make rafts for trade & fishing because bamboo is light, hollow & can easily float on water.

Density increases with cooling & decreases with heating. Because when something is heated, its particles tend to move away, this causes a change in volume. & As density is just $\frac{Mass}{Vol}$ so when volume increases, density decreases. This concept makes the hot air balloon work as hot air is less dense & moves up.

~~# Pressure can increases a gases density.~~

Pressure increases gas's density a lot as they have a lot of space b/w their particles to compress. But, pressure has minimal effects on liquid & no effect on solid.

Ice floats on water, even though both are water. Water is the most dense at 4°C but ice freezes well below 0°C, so when water freezes particles arrange themselves in a way that they take up more space. This process is called expansion. Expansion increases the volume & decreases density, that's why ice floats on water.

This floating of ice forms a layer on water & keep water warm enough for marine life to survive.

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Asima Chatterjee was an Indian Scientist who worked on developing anti-malarial & anti-epileptic drug. She earned a doctorate of Science. First woman to recieve Shanti Swarup Bhatnagar Award. Recieved Padma Bhushan. She used extract & isolate important compounds from medicinal plant.